

Clinical and Translational Science Sustainability: Overcoming Integration Issues between Electronic Health Records (EHR) and Clinical Research Data Management Systems “Separate but Equal”

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Abstract

The use of health information technology (HIT) is growing rapidly for patient care systems required to test, diagnose and treat patients, and to bill for these services. Today's Electronic Health Record (EHR) systems are a response to this pressure, enabling feature rich computer-assisted decisions and communication. And even though EHR benefits dramatically outweigh the costs, required investments are nonetheless significant. Continuing to invest in HIT at a revolutionary rate is unsustainable given institutional financial constraints and continuing reimbursement cuts. Future improvements must come from new treatments, test methods, drugs and devices – from research. But data management information systems for clinical research receive less funding than patient care systems, and in less coherent ways. It is easy to imagine using the high cost, patient-based EHRs for clinical research data management, and thus accelerate the speed of translating new medical discoveries into standard practice. But taking this step requires thoughtful planning to overcome significant technology, legal/regulatory, policy, process, and administrative issues.

Keywords:

medical informatics, biomedical research, clinical research, information systems, computerized medical record systems

Introduction

Although the challenge of defining an Electronic Health Record (EHR) with multinational agreement has vexed experts around the globe, several professional associations are now reaching consensus on a basic conceptual elements.[1-3] An EHR uses computer systems to enter, store and retrieve patient medical records with accessibility by various authorized individuals. Additionally and more importantly, an EHR's extended functionality and interoperability bring significant benefits to users, healthcare providers and payers across multiple EHR systems and geographic locations (the Integrated Care EHR). Primary benefits come from faster and better decisions and reduced errors, thus improving care and saving lives. But looking more carefully at the financial picture, EHR adoption rates internationally vary based on country, care setting and pro-

vider specialization [4-5], typically anywhere between 15-85%, compared to the U.S. norm of about 25% for large hospitals[6]. While costs for nationwide implementation are well known where full adoption is nearly complete (e.g., Sweden, U.K.), other countries can only estimate the full financial requirements of such large scale projects, albeit with increasing accuracy. For the U.S., with the world's largest annual healthcare expenditures of around US\$2 trillion, the cost projection for full national implementation of an ambulatory and inpatient EHR over 15 years is \$114.6 billion.[7] This same report also estimates a 548% return on these investments, with savings of nearly \$627.5B. Clearly the global trend to adopt nationwide EHRs is underway and will continue into the next decade. Unfortunately adoption has been slow in countries with market-based economies and privatized healthcare since financial models don't show the anticipated savings actually being accrued by the organizations that make the investments. Some large scale adopters have created new associations to encourage collaboration and financial intermediation, like the Good European Health Record project [8], or the not-for-profit company behind Australia's National E-Health Transition Authority (NEHTA), whose mission is to deliver a secure, interoperable nationwide e-health environment.[9] Such mechanisms may inevitably be required in the U.S. since projected HIT spending (Figure 1), currently less than 1% of overall healthcare cost, is expected to climb steadily (averaged estimate from CMS, IDC, Gartner, Frost & Sullivan, BCC Research, Datamonitor, Kalorama).

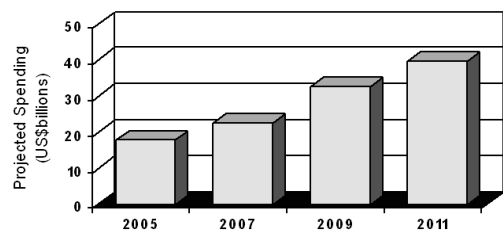


Figure 1 – Healthcare IT spending estimate for US (\$billions)

U.K. national EHR implementation costs

To highlight the scale of coordination and funding required for national EHR implementation, The Lewin Group [10] summarized the U.K. National Health Service (NHS) in 2005 as:

“...the largest organization in Europe and the third largest employer in the world, with more than 1.3 million workers. The agency is generally regarded as an efficient provider of high-quality health care to a large population. The state finances the NHS mainly through general taxes, administers the agency and makes executive decisions. Approximately 60% of U.K. primary practitioners use electronic health records. Recently, the NHS embarked on the world's largest civilian IT project, a plan to spend \$11 billion on a national HIT system to replace existing local systems and paper medical records.”

More recent reports have shown the size of spending in the U.K. may surpass US\$20 billion before final goals are met, due partly from incorrect initial estimates and inflation, but also likely from scope creep for secondary uses of data.

Comparison with U.S. EHR funding

While the U.S. president has called for a national EHR to be available for most Americans by 2014, the necessary funding levels shown to be required in the U.K. haven't been provided thus far. And the FY2007 budget proposal recently submitted to Congress included only a request of \$169M for EHR implementation, and \$116M for the Office of the National Coordinator for Health Information Technology to coordinate and plan a national strategy and initiatives. The initial increase for the National Institutes of Health (NIH) in 2006, which saw a doubling of its budget over the past five years to approximately \$28B, was virtually zero [11], although this was increased slightly by the NIH Reauthorization Act passed in December. But nearly two thirds of the NIH budget goes toward basic research, with only the remaining one third going to patient-oriented clinical research versus patient care; i.e., generally not for health information system infrastructure costs such as an EHR. Thus meeting the necessary expense for nationwide EHR implementation in the U.S. with such limited funding is unrealistic, with current efforts simply unsustainable.

Clinical research and its role in healthcare

Clinical research (CR) is defined by the NIH as human subject research involving an investigator that is patient-oriented including the study of – mechanisms of disease, therapeutic interventions, clinical trials (CT), and the development of new technologies. They may also involve epidemiologic and behavioral studies, outcomes research, be retrospective or prospective, utilize standard-of-care control groups or placebo comparisons, may randomize carefully selected samples of study subjects from larger groups or investigate specifically selected individuals because of their health circumstances.[12] Human subject research must follow strict ethical requirements that have been agreed to by well known international collaborations such as the World Medical Association Declaration of Hel-

sinki, as well as regulatory requirements within, and sometimes between, national borders. Regulatory obligations typically mandate an independent Institutional Review Board (IRB) that evaluates all requests for research and ensures necessary policies and procedures are complied with. And since even large, well-known institutions can be involved with investigators who at times act inappropriately or who are not sensitive enough to study subjects special needs, especially for vulnerable populations (e.g., children, elderly, etc.), new positions and procedures have been created to independently review and endorse an institution's research enterprise and commitment to research integrity.[13]

EHRs and healthcare in general can be looked upon as vertical space in an overall strategic framework for biomedical science. The knowledge transfer that occurs from translation of scientific discovery in bench laboratories, to animal research, *in silico* research, pilot studies in humans, and all the way through large scale, multi-center, global studies into standard practice by healthcare providers was described by Sung et al [14] as the T1 translational block horizontally across these domains. Clinical research is the scientific domain prior to healthcare delivery, and upon which new standards of care depend. The need to improve the productivity and cost effectiveness of CR processes and data management systems is similar to healthcare, although on a smaller scale. The need to more tightly couple CR to healthcare, and move new discoveries forward across these domains quickly and inexpensively is also important.

Like healthcare costs, various reports [15-16] have studied the unsustainability of regulated clinical trials in particular as a subset of CR for drug and device discoveries to meet growing demands to control future healthcare problems and changing patient population demographics given the large expense and years required to get final regulatory approvals.

EHR/Clinical research integration

EHRs and CR data management systems (Figure 2) can exist as anything from independent, parallel technology platforms,

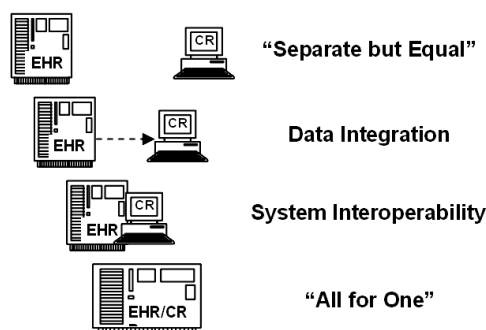


Figure 2 – EHR/CR Integration Models

to the CR system receiving a one-way data feed from the EHR, to some degree of two-way interoperability between the separate systems, to being completely integrated into

one system. This fourth alternative seems to be mostly theory at this point considering major EHR vendor products. An informal survey conducted during the 2006 HIMSS conference found only 20% with CR capabilities in current EHRs. This finding was similar to a more formal study by the American Academy of Neurology that found only 28.5% of mid-sized EHR vendors to be capable of conducting any level of CR in their current systems.[17] Usually a central IT organization governs the EHR. It also has certain budgetary or systems responsibility for CR systems, though whenever this is the case, the CR mission is almost always a very small part of the overall IT focus and funding.[18] In fact this study showed 60% of academic health centers do not even have a clear picture of how much is spent on IT for CR, and 49% have no central HIT-CR governance structure or cross-representation between these personnel. Virtually none have a strategic plan for CR. And while various attempts have been made to begin experiments to integrate EHRs with CR systems, or even to fully conduct a CR study completely within the EHR, most have been very small or ineffective due to the overwhelming issues that must be faced to make these connections work and leverage the significant investment in EHRs. Even demanding these connections through legislation in a national setting has proven futile.[19]

Issues in EHR/CR integration

As stated earlier there are many types of CR studies. All may not be appropriate to conduct directly within an EHR or via an integrated EHR/CR system, but it may be easily possible for some. The following are general categories of issues surrounding EHR/CR integration.

Technology

Most EHR products in the U.S. (other than for an OpenEHR) are proprietary technologies with vendor controlled code bases requiring certain operating platforms and network architectures. Interoperability with other information systems may be possible in some cases, but underlying data models, structure, format, vocabulary and transmission standards are certain to inhibit connectivity or data transfer without significant work.

Current FDA requirements for clinical trials of investigational drugs and devices specify that the data management system comply with system and software validation.[20-21] The steps to comply with these requirements are typically very expensive, and must be repeated whenever changes are made to the technology, which is unacceptable in a large EHR.

Legal/Regulatory

In the U.S. (similar to other countries), each patient obtaining medical treatment must be shown a Notice of Privacy Practice (NoPP) that informs them of their legal rights under the Healthcare Insurance Portability and Accountability Act (HIPAA). The notice requires “plain language”, and a statement that information collected on the patients for standard care may be used for research (operational and outcomes) in addition to treatment, payment and quality purposes.[22] The use of patient data for future,

unspecified research is *not* permitted. An assessment of the NoPP from the twelve winners of the new NIH Clinical and Translational Science Award (CTSA) for 2006 [23] indicates an average Flesch-Kincaid Grade Level of 14.3. Certainly most hospital patients are not college level readers. So even though the CTSA winners have been recognized for outstanding abilities to perform CR and to translate new science into the patient care domain, there will most likely need to be changes made in how they inform patients that research may be conducted on their data (possibly in the EHR) as normal care.

Conversely, whether a patient or not in a hospital, any person who is going to participate in prospective CR is required by regulation to be fully informed of all aspects of the study prior to making a decision to participate, and not to be coerced in any way, especially by any person in a position of authority (as is a physician-investigator when discussing a research study they are the principal investigator on with their patient). Typical informed consent forms for CR are 10-20 pages in length and can be quite complex [24] even though they should be written at less than a high-school reading level.[25]

Policy

A hallmark of an IRB is its ability to independently make ethical decisions that balance institutional interests in research against protecting human subjects involved in studies. Like criticism of the FDA because it receives money from drug companies as part of the drug approval process, the appearance of improper behavior or the potential conflict of interest is often enough to cause problems. At what point does an IRB, which may already be closely aligned with or supported by an institution, become too close and biased?

Revenues from research, whether federally funded or from industry sponsored sources generate a significant amount of income at large institutions. At what point does strict compliance with someone’s interpretation of an SOP, or an unusual pattern of missing values, or delayed serious adverse event reporting require the institution to suspend or end a study (and revenue)?

How is time seeing patients as part of normal care accounted for by doctors when patients are also enrolled in a research study in the same visit? How are the costs separated, or insurance companies fully informed that the visit is for “research”?

Process

Fundamentally all CR processes are very similar, yet different enough that it has been easier to build required data management system from scratch every time a new study comes along. And since most funding for CR studies has been provided independently with little to no regard for centralized or reusable systems, investigators either received funding to build everything they needed from scratch every time, or they received little to no money for data management needs at all and thus had to cut corners in order to conduct an under-funded study in whatever way was possible within their institution. Standards for CR

(Figure 3) that improve operational consistency now exist however, especially where countries fund national EHRs.

Physicians are thus under increasing pressure to follow newly prescribed processes, especially in the case when an EHR is in place; often having 10 minutes or less to interact with patients. How can they provide the necessary additional time required to inform a patient fully about a new research study patients are eligible to participate in? Should they delegate?

Likewise, research studies have historically been statistically designed around strict visit schedules and data collection methods in order to ensure validity and significance. In many cases, these study schedules are not the standard-of-care that a physician would normally provide for their patients. So patients need

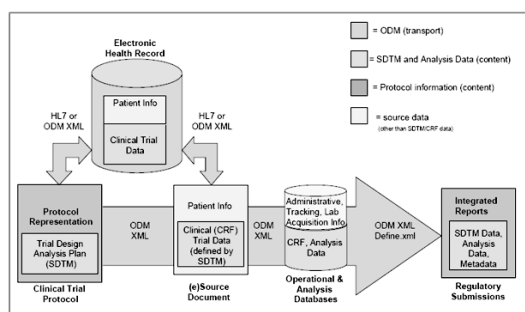


Figure 3 – Clinical Data Flow Using CDISC Std. [26]

to either come in more frequently than is normally necessary, or visit only when they are scheduled regardless of how they feel. Both situations are problematic.

Administrative

Security (i.e., confidentiality, integrity, accessibility, etc.) of a research subject's data is at least as important as a patient's data. Yet there are different rules that govern how data is managed and who can access what. Especially if CR data is collected as part of an EHR; consent forms often limit who can see study subject data to only people directly involved with the research, as opposed to anyone in the institution with general access to the EHR who is allowed to see patient records. How can this distinction be readily defined, and who will have the real-time responsibility to enter such permissions in an EHR as security administrator? Finally, healthcare data breaches are more commonplace than people want to know.[27]

EHR systems typically have well defined workflow components designed into the data collection screens for ease of use. Will new fields or special screens need to be designed for all research trials to enable complete and timely collection of relevant study variables (eCRF)? Will study subjects have the legal right to withdraw from research participation at any time for any reason without repercussion to their normal care, or to prohibit future use of their data for research purposes?

Recommendations for progress

The consensus from leading U.S. academic medical research centers at the Clinical Research Forum meeting in April 2007, as well as new dialog on this topic held at the 2007 HIMSS meeting with the EHR Vendor Association and pharmaceutical companies was that there are many issues to overcome, but none are impossible. Participation of all stakeholders is necessary to create sustainable success. There is no single "leader" or starting point to EHR/CR integration. Pick any spot relative to your unique environmental needs and just get started!

Regarding technology, if you already have an EHR, begin a dialog with your vendor on inclusion of CR into their functional design. Require this from them in the future. If you don't have an EHR, become involved in national associations that are beginning to hold these public discussions (e.g., AMIA) with mixed stakeholder groups. Require vendor involvement and compliance with nationally recognized standards. Pressure publicly elected officials regarding regulatory agencies to accelerate changes in outdated rules and guidelines in support of electronic source data collection and transmission.

Challenge legislators through efforts with national associations to harmonize the legal framework around patients and study subjects, thus lowering the burden associated with CR translational science. Rethink and revise plans and materials for patient and community awareness, and their involvement in CR including NoPP and informed consent document creation.

Form or re-energize institutional committees defining responsible research conduct, research integrity, or conflicts of interest. Establish policies for limits to IRB involvement in hospital operational research; budgeting and accounting for research activities and investigational staff effort reporting.

Investigate, then adopt or adapt a CR process model. Benchmark best practices, including sharing your work with others. And finally, continuously improve and regularly test your security model for patient and research data. It's never secure.

Conclusion

EHRs implemented on a national scale are inevitable over the next decade, but there must be significant financial support and coordinated collaboration among many stakeholders to succeed and overcome the unsustainability of sub-optimized approaches. Likewise CR data management systems have diminishing chances of existing on their own as separate source records because of the many issues identified in this paper. Integration of the EHR with CR systems, or better, enabling CR to be completely carried out within the EHR system itself, will take foresight and redesign work between central HIT staff, CR informaticists, and commercial and open source partners. In addition to the financial and quality-of-care benefits, future research will be able to leverage clinical data-CR repositories that will become available from regional and

national networks. Study subject enrollment may quickly increase and open a way to explore rare “orphan” and pediatric diseases that would have otherwise been financially or statistically unattainable.

Acknowledgments

Thanks to Drs. Anil Jain, Mary Partin, Michael Kattan and others at Cleveland Clinic; Dr. Pam Davis at CASE; Michael Lin of Mayo Clinic; Drs. Elaine Collier and Peter Highnam at the NIH/NCRR; Rob Schuff at Oregon Health Sciences University; Dr. Paul Harris at Vanderbilt; Dr. Michael Kahn at University of Colorado Health Sciences Center; and Dr. Don Detmer from AMIA for their ongoing encouragement, challenge and support.

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